

A NOTE ON A CATEGORY OF NOISY CHANNELS

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27/8/2026

We want to define a monoidal category where morphisms are noisy channels. Intuitively, they are composed of a deterministic mapping $f: A \rightarrow B$ and a source of stochastic noise $\sigma: U \rightarrow B$ that depends on a global parameter U ; the output is the sum of the output of f and the noise; so, we need a monoid structure on each object and a copy-discard structure to share parameters. The goal is to generalise the Gauss construction [Gritz(2020)] of affine maps with gaussian noise.

DEFINITION

Consider:

- $\mathcal{C}$ a copy-discard category;
- $\mathcal{Z} \subseteq \mathcal{C}$, a wide monoidal subcategory of total and deterministic morphisms;
- $\mathcal{D} \subseteq \mathcal{C}$, a wide monoidal subcategory of $\mathcal{C}$ containing $\mathcal{Z}$;
- $\text{Mon}(\mathcal{Z}, \mathcal{D})$, the category of monoids in $\mathcal{Z}$ with monoid morphisms in $\mathcal{D}$;
- $U \in \text{obj } \mathcal{C}$, an object of $\mathcal{C}$.

Define a category $\text{Noise}(\mathcal{Z}, \mathcal{D}, \mathcal{C})$ with the same objects as $\text{Mon}(\mathcal{Z}, \mathcal{D})$ and where morphisms $(f, \sigma): (A, m_A, u_A) \rightarrow (B, m_B, u_B)$ are pairs of a monoid morphism in $\mathcal{D}$, $f: (A, m_A, u_A) \rightarrow (B, m_B, u_B)$, and a morphism, $\sigma: U \rightarrow B$, in $\mathcal{C}$.

LEMMA

$\text{Mon}(\mathcal{Z}, \mathcal{D})$ is a copy-discard category.

PROOF

It's a known fact. copy and discard are monoid morphisms because we imposed that multiplications and units be total and deterministic.

PROPOSITION

The construction $\text{Noise}_0(\mathbb{Z}, \mathbb{D}, \mathcal{C})$ defines a symmetric monoidal category.

PROOF

We prove that it is a category.

Compositions.

For two morphisms, $(f, \sigma): A \rightarrow B$ and $(g, \tau): B \rightarrow C$, their composition is defined as $(f, \sigma); (g, \tau) := (f; g, \underbrace{\sigma \circ \tau}_{\tau \circ \sigma})$.

Identities.

For an object (A, m_A, u_A) , its identity is defined as $\text{id}_{(A, m_A, u_A)} := (\text{id}_A, \rightarrow \circ -)$.

Associativity.

$$\begin{aligned} & ((f, \sigma); (g, \tau)); (h, \rho) \\ &= (f; g, \underbrace{\sigma \circ \tau}_{\tau \circ \sigma}); (h, \rho) \\ &= ((f; g); h, \underbrace{\underbrace{\sigma \circ \tau}_{\tau \circ \sigma} \circ \rho}_{\rho \circ (\sigma \circ \tau)}) \\ &= ((f; g); h, \underbrace{\underbrace{\sigma \circ \tau \circ \rho}_{\tau \circ (\sigma \circ \rho)}}_{\rho \circ (\sigma \circ \tau)}) \\ &= (f; (g; h), \underbrace{\underbrace{\sigma \circ \tau \circ \rho}_{\tau \circ (\sigma \circ \rho)}}_{\rho \circ (\sigma \circ \tau)}) \\ &= (f, \sigma); (g; h, \underbrace{\tau \circ \rho}_{\rho \circ \tau}) \\ &= (f, \sigma); ((g, \tau); (h, \rho)) \end{aligned}$$

Unitality.

$$\begin{aligned} & (f, \sigma); (\text{id}_B, \rightarrow \circ -) \\ &= (f; \text{id}_B, \underbrace{\sigma \circ \text{id}_B}_{\text{id}_B \circ \sigma}) \\ &= (f, \sigma) \end{aligned}$$

$$\begin{aligned} & (\text{id}_A, \rightarrow \circ -); (f, \sigma) \\ &= (\text{id}_A; f, \underbrace{\text{id}_A \circ \sigma}_{\sigma \circ \text{id}_A}) \\ &= (f, \underbrace{\text{id}_A \circ \sigma}_{\sigma \circ \text{id}_A}) \\ &= (f, \sigma) \end{aligned}$$

We prove that it's a SMC.

On objects, the monoidal product is the product of monoids in a symmetric monoidal category.

For two morphisms, $(f, \sigma): A \rightarrow B$ and $(f', \sigma'): A' \rightarrow B'$, their monoidal product is componentwise,

$$(f, \sigma) \otimes (f', \sigma') := (f \otimes f', \underbrace{\sigma \otimes \sigma'}_{\sigma' \otimes \sigma}).$$

Functoriality in each variable is easy to check.

Interchange law.

$$\begin{aligned} & ((f, \sigma) \otimes (f', \sigma')); ((g, \tau) \otimes (g', \tau')) \\ &= (f \otimes f', \underbrace{\sigma \otimes \sigma'}_{\sigma' \otimes \sigma}); (g \otimes g', \underbrace{\tau \otimes \tau'}_{\tau' \otimes \tau}) \\ &= ((f \otimes f'); (g \otimes g'), \underbrace{\underbrace{\sigma \otimes \tau}_{\tau \otimes \sigma} \otimes \underbrace{\sigma' \otimes \tau'}_{\tau' \otimes \sigma'}}_{\underbrace{\sigma \otimes \tau' \otimes \sigma' \otimes \tau}_{\sigma' \otimes \sigma \otimes \tau' \otimes \tau}}) \\ &= ((f; g) \otimes (f'; g'), \underbrace{\underbrace{\sigma \otimes \tau}_{\tau \otimes \sigma} \otimes \underbrace{\sigma' \otimes \tau'}_{\tau' \otimes \sigma'}}_{\underbrace{\sigma \otimes \tau' \otimes \sigma' \otimes \tau}_{\sigma' \otimes \sigma \otimes \tau' \otimes \tau}}) \\ &= (f; g, \underbrace{\sigma \otimes \tau}_{\tau \otimes \sigma}); (f'; g', \underbrace{\sigma' \otimes \tau'}_{\tau' \otimes \sigma'}) \\ &= ((f, \sigma); (g, \tau)) \otimes ((f', \sigma'); (g', \tau')) \end{aligned}$$

Associators and unitors come from $\mathcal{C}$ and satisfy $\square$ and $\triangleright$ as a consequence.

Same for symmetries and their equations. $\square$

NOTE. This result only uses that the first components of the morphisms are monoid morphisms and the copy-discard structure of $\mathcal{C}$; it does not use the copy-discard structure of $\text{Mon}(\mathbb{Z}, \mathbb{D})$.

LEMMA

There is an identity on objects symmetric monoidal functor,

$E: \text{Mon}(\mathcal{Z}, \mathcal{D}) \rightarrow \text{Noise}_v(\mathcal{Z}, \mathcal{D}, \mathcal{C})$,
defined by $E(f) := (f, \rightarrow \circ -)$; it maps monoid morphisms to their corresponding noiseless channels.

PROOF

We only need to check functoriality.

$$\begin{aligned} E(f); E(g) &:= (f, \rightarrow \circ -); (g, \rightarrow \circ -) \\ &:= (f; g, \rightarrow \circ \circ -) \\ &= (f; g, \rightarrow \circ -) \\ &= (f; g, \rightarrow \circ -) \\ &:= E(f; g) \end{aligned}$$

$$\begin{aligned} E(\text{id}_{(A, m_A, u_A)}) &:= (\text{id}_A, \rightarrow \circ -) \\ &:= \text{id}_{(A, m_A, u_A)} \end{aligned}$$

□

COROLLARY

$\text{Noise}_v(\mathcal{Z}, \mathcal{D}, \mathcal{C})$ is a copy-discard category.

REMARK

The copy-discard category $\text{Noise}_v(\mathcal{Z}, \mathcal{D}, \mathcal{C})$ doesn't seem to have conditionals in general. What's special about Gauss that makes it with conditionals?

REMARK

One may want to allow the noise to depend on inputs as well, i.e.

$\sigma: A \otimes U \rightarrow B$ in $\mathcal{C}$, with composition

$$\boxed{f} \circ \boxed{g} = \text{diagram with } f, g, \text{ and } \sigma \text{ boxes and wires};$$

for this, we would need f and g deterministic, but the rest should work the same.